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GATE 2010 EE, 52^{nd} Question Analysis

Question 52

Statement for Linked Answer Questions: 52 & 53

The following Karnaugh map represents a function F.

F \ X	00	01	11	10
0	1	1	1	0
1	0	0	1	0

52. A minimized form of the function *F* is

(A) $F = \overline{X}Y + YZ$ (B) $F = \overline{X}\overline{Y} + YZ$ (C) $F = \overline{X}\overline{Y} + Y\overline{Z}$ (D) $F = \overline{X}\overline{Y} + \overline{Y}Z$

Question Analysis:

- From above kmap, the output F is as follows
- $F = \bar{X}.\bar{Y} + Y.Z$
- The Truth table for above expressions is as follows

The Truth Table

X	Y	Z	$F = \bar{X}.\bar{Y} + Y.Z$
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Hardware Implementation

The above problem is implemented and tested in hardware using Arduino UNO board. Here we implemented a FSM using the 7474 IC and blinked the LED as per truth table and verified the expression.

Required Components & Pin Connections

S.No	**Component**
1	Arduino Uno Board
2	Breadboard
3	7474 IC (2)
4	LEDs (1)
5	7447 IC (1)
6	Seven segment (1)
7	Resistors: 220Ω (2)
8	Jumper Wires
9	USB Cable

Component	**Arduino Pin**
Input Z (7474-1 Q1)	Digital 2
Input Y (7474-1 Q2)	Digital 3
Input X (7474-2 Q1)	Digital 4
Output A (7447 A)	Digital 8
Output B (7447 B)	Digital 9
Output C (7447 C)	Digital 10
Output D (7447 D)	Digital 11
Output F (LED)	Digital 7
Output clk (7474 clk)	Digital 13
GND	GND
VCC	5V

Logic Description

- Let initialize inputs $X = 0, Y = 0, Z = 0$
- The output expressions from state transition table are reduced using KMAP and are as follows

- $A = \bar{Z}$
- $B = \bar{Y}.Z + Y.\bar{Z}.$
- $C = X.\bar{Y} + X.\bar{Z} + \bar{X}.Y.Z$
- $d = 0$
- $F = \bar{x}.\bar{z} + X.Y$

State Transition Table

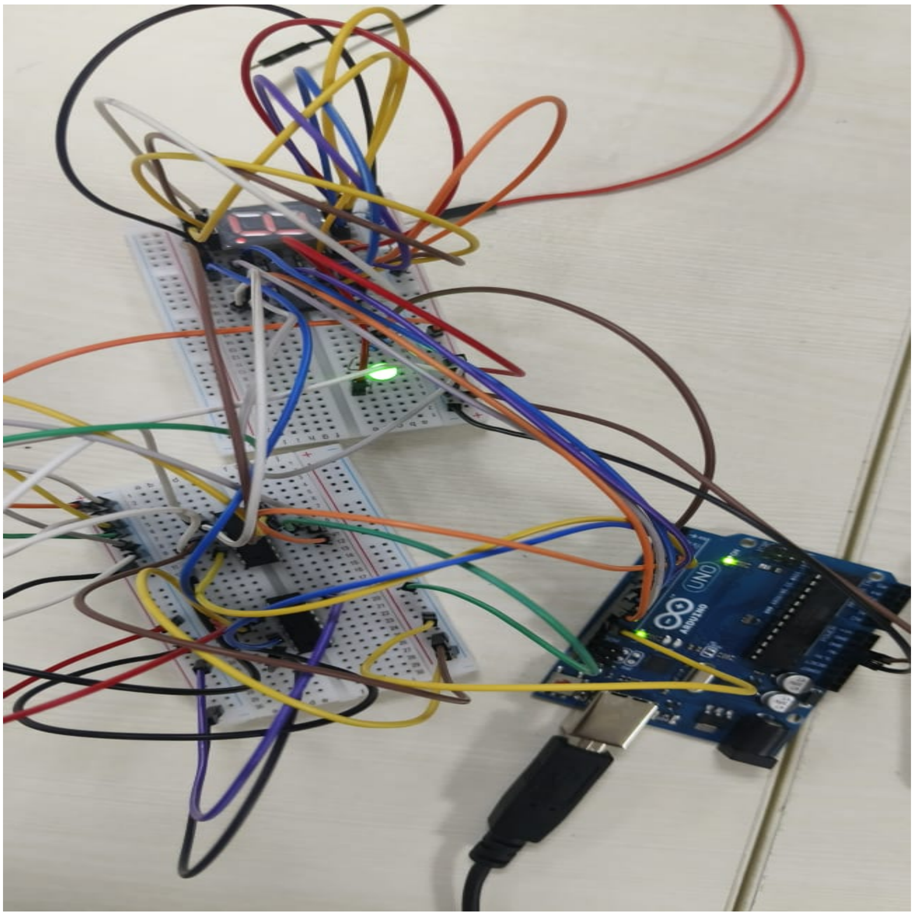
Present State			Next State				Output
X	Y	Z	D	C	B	A	F
0	0	0	x	0	0	1	1
0	0	1	x	0	1	0	0
0	1	0	x	0	1	1	1
0	1	1	x	1	0	0	0
1	0	0	x	1	0	1	0
1	0	1	x	1	1	0	0
1	1	0	x	1	1	1	1
1	1	1	x	0	0	0	1

Code Uploading Steps

1. Create a fsm folder and write Makefile
2. Write The code in main.c
3. Run the command "make". It will compile the code and creates .hex file
4. Copy the .hex file to ArduinoDriod folder
5. connect the Arduino UNO to mobile with OTG cable
6. Upload the hex file using "upload precomplied" option
7. Observe the ouput and verify the expression

Experimental Truth Table

X	Y	Z	F (LED Output)
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1



Conclusion

- From Experimental Truth Table F will be 1 when $C = 1$.
- This matches option (D) from the original GATE question.
- The hardware experiment confirms the circuit's theoretical logic.